

FOUNDATION OF COMPUTER SCIENCE (MTCS101)

UNIT-I: DATA STRUCTURE

List, Stack, Queue, Tree, Hash Table, Graph, Search and Sorting Algorithms.

UNIT-II: OPERATING SYSTEM

Scheduling Algorithm, Synchronization Technique, Paging and Segmentation, Virtual Memory.

UNIT-III: AUTOMATA THEORY

Finite Automata, Regular Expression, Context Free Grammar, Push Down Automata, Turing Machine, P and NP Class.

UNIT-IV: DATABASE SYSTEM

Concepts and Architecture; Data Model; Normalization; SQL Advanced Transaction Processing, Deadlock and Concurrency Control; Object Oriented and Object Relational Databases: Parallel and Distributed Databases; Backup and Recovery Concepts, Emerging Database Technologies.

REFERENCES:

- 1.Hopcroft & Ullman, "Introduction to Automata Theory, Languages, and Computation", Narosa Publishing House.
- 2.Aho, Hopcroft and Ullman, "Data Structures and Algorithms", Addison Wesley.
- 3.Dhamdhare, "Operating Systems", Tata McGraw Hill.
- 4.Aho, Ullman and Sethi, "Compiler Design", Addison Wesley.
- 5.Ramez Elmasri, Shamkant Navathe: Fundamentals of Database Systems, Fifth Edition, Pearson Education, 2007.
- 6.Raghu Ramakrishnan, Johannes Gehrke : Database Management Systems, Tata McGraw-Hill.
7. Alexis Leon, Mathews Leon, "Database Management Systems
8. C.J. Date : An Introduction to Database Systems, Eighth Edition, Pearson Education.
9. Abraham Silberschatz, Henry F. Korth, S.Sudarshan: Database System Concepts, Tata McGraw-Hill.

ADVANCED ALGORITHM (MTCS102)

Algorithm Fundamentals: Basic Concept, Analysis of Algorithm, Growth of Functions, Master's Theorem.

Analysis of sorting Algorithms: Overview, Merge sort, Quick sort, Heap sort, radix sort.

Advance Data Structure: Red-Black Trees, B/B+ Trees.

Parallel Algorithm: Performance Measures of Parallel Algorithms, Parallel Merging/Sorting Algorithms on CREW/EREW, Parallel searching algorithms.

Advance Design and Analysis Techniques: Dynamic Programming, Greedy Algorithms, Branch and bound, Back Tracking.

Graph Algorithm: DFS and BFS algorithm.

NP Complete Problem

SOFTWARE PROCESS & MANAGEMENT (MTCS012)

UNIT-I:

DEVELOPMENT LIFE CYCLE PROCESSES

Overview of software development life cycle – introduction to processes – Personal Software Process (PSP) – Team software process (TSP) – Unified processes – agile processes – choosing the right process Tutorial: Software development using PSP

UNIT-II:

REQUIREMENTS MANAGEMENT

Functional requirements and quality attributes – elicitation techniques – Quality Attribute Workshops (QAW) – analysis, prioritization, and tradeoff – Architecture Centric Development Method (ACDM) – requirements documentation and specification – change management – traceability of requirements Tutorial: Conduct QAW, elicit, analyze, prioritize, and document requirements using ACDM

UNIT-III:

ESTIMATION, PLANNING, AND TRACKING

Identifying and prioritizing risks – risk mitigation plans – estimation techniques – use case points – function points – COCOMO II – topdown estimation – bottomup estimation – work breakdown structure – macro and micro plans – planning poker – wideband delphi – documenting the plan – tracking the plan – earned value method (EVM) Tutorial: Estimation, planning, and tracking exercises

UNIT-IV:

CONFIGURATION AND QUALITY MANAGEMENT

identifying artifacts to be configured – naming conventions and version control – configuration control – quality assurance techniques – peer reviews – Fegan inspection – unit, integration, system, and acceptance testing – test data and test cases – bug tracking – causal analysis Tutorial: version control exercises, development of test cases, causal analysis of defects

UNIT-V:

SOFTWARE PROCESS DEFINITION AND MANAGEMENT

Process elements – process architecture – relationship between elements – process modeling – process definition techniques – ETVX (entrytaskvalidationexit) – process baselining – process assessment and improvement – CMMI – Six Sigma Tutorial: process measurement exercises, process definition using ETVX

REFERENCES:

1. Pankaj Jalote, “Software Project Management in Practice”, Pearson, 2002.

NEURAL NETWORKS (MTCS026)

UNIT-1:

Basics of ANN: Models to Neuron; Basic learning laws. Activation and Synaptic Dynamics: Activation dynamics models; Synaptic dynamics models; Stability and Convergence.

UNIT-II:

Analysis of Feed forward Neural Networks: Linear associative networks for pattern association; Single layer and Multilayer Perception network for pattern classification; Multilayer feed forward neural networks for pattern mapping.

UNIT-III:

Analysis of Feedback Neural Networks: Linear auto associative networks; Hopfield model for pattern storage; stochastic networks; Boltzmann machine for pattern environment storage.

UNIT-IV:

Competitive Learning Neural Networks: Basic competitive learning laws; Analysis of pattern clustering networks; Analysis of self-organizing feature mapping networks.

UNIT-V:

Applications of ANN: Pattern classification problems; Optimization; Control.

REFERENCES:

1. J.A. Anderson, An Introduction to Neural Networks, MIT
2. Hagen Demuth Beale, Neural Network Design, Cengage Learning
3. Laurene V. Fausett, "Fundamentals of Neural Networks : Architectures, Algorithms and Applications", Pearson India
4. Kosko, Neural Network and Fuzzy Sets, PHI